

07/21/25 - Video 2.2 - Masters Theorem Decreasing Functions

From the previous videos, we know that

1. $T(n) = T(n-1) + 1$ is $O(n)$
2. $T(n) = T(n-1) + n$ is $O(n^2)$
3. $T(n) = T(n-1) + \log n$ is $O(n \log n)$
4. $T(n) = 2T(n-1) + 1$ is $O(2^n)$

- The pattern for 1-3 is simple, whatever's after the $T(n-1)$ is multiplied by n .
- For the $2T(n-1)$ whatever's after that, is multiplied by 2^n .

↳ Using #4, this means $T(n) = 3T(n-1) + 1$ is $O(3^n)$

And using our previous patterns from before, we can say that $T(n) = 2T(n-1) + n$ is now $O(n 2^n)$.

General Form/Rules

$$T(n) = aT(n-b) + f(n)$$

O or Θ

- $a > 0$
- $b > 0$
- $f(n) = O(n^k)$ where $k \geq 0$

1) • if $a = 1$, then $O(n \cdot f(n))$

2) • if $a > 1$, then $O(f(n) \cdot a^n)$

↳ if $b > 1$, the formula then changes
 $O(a^{n/b} \cdot f(n))$

3) • if $a < 1$, then $O(f(n))$

Suppose we have the following code

```

Algorithm Test(int n) → T(n)
{
  if (n > 1) → irrelevant
  {
    print(n); → 1
    Test(n/2); → T(n/2)
  }
}
    
```

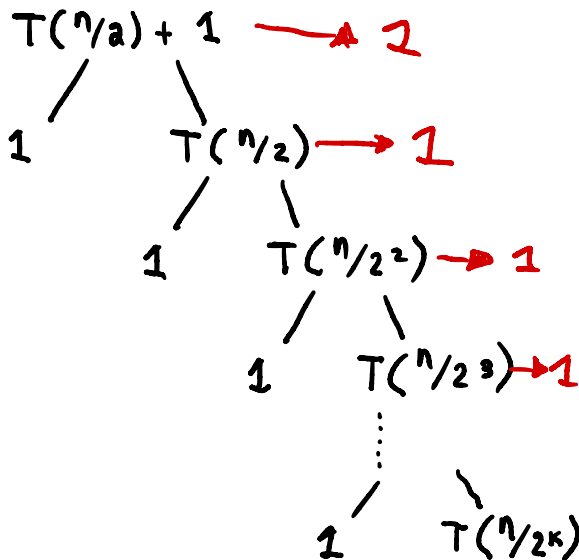
Given an input n , a function can decrease its value by

$n-1$
 $n/2$
 $\sqrt{n}$

Full recurrence relation ↓

$$T(n) = \begin{cases} 1 & , n=1 \\ T(n/2) + 1 & , n>1 \end{cases}$$

Dealing with $T(n) = T(n/2) + 1$



• now to solve this, we make a recurrence tree, as we keep dividing by 2, we see that its $T(n/2) \rightarrow T(n/4) \rightarrow T(n/8) \rightarrow T(n/16)$ etc etc.

• And based on our full recurrence, we want to reach 1, so $n/2^k = 1$.

• Now find the **TIME COMPLEXITY**, so

$$\frac{n}{2^k} = 1 \rightarrow n = 2^k \rightarrow k = \log n$$

so $\boxed{O(\log n)}$ ✓

Substitution Method for

$$T(n) = \underbrace{T(n/2)}_{?} + 1$$

$$\begin{aligned} & \rightarrow T(n) = T(n/2) + 1 \\ & T(n/2) = T(n/4) + 1 \end{aligned}$$

$$T(n) = [T(n/4) + 1] + 1$$

$$T(n) = \underbrace{T(n/4)}_{?} + 2$$

$$\rightarrow T(n/4) = T(n/8) + 1$$

$$T(n) = [T(n/8) + 1] + 2$$

$$T(n) = T(n/8) + 3$$

$$T(n) = T(n/2^3) + 3$$

$$\vdots$$

$$T(n) = T(n/2^k) + k$$

From our full recurrence relation,
we know when $n=1$, it's a
constant.

$\therefore$ ASSUME

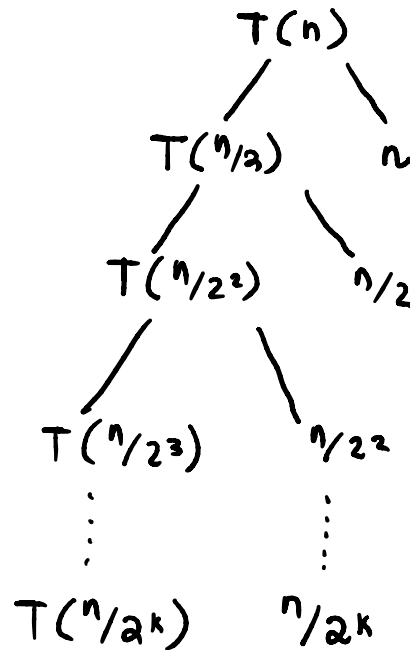
$$\frac{n}{2^k} = 1$$

$$\begin{aligned} n &= 2^k \\ k &= \log n \end{aligned}$$

$$\boxed{O(\log n)} \checkmark$$

Suppose we have the following relation

$$T(n) = \begin{cases} 1 & n = 1 \\ T(n/2) + n & n > 1 \end{cases}$$



$$T(n) = \overbrace{n + n/2 + n/2^2 + \dots + n/2^k}$$

$$T(n) = n \left[1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^k} \right]$$

$$= n \sum_{i=0}^k \frac{1}{2^i} = 1$$

$$= n \times 1$$

$$T(n) = \boxed{O(n)}$$

Substitution Method

$$T(n) = \underbrace{T(n/a)}_{?} + n$$

$$? \rightarrow T(n) = T(n/2) + n$$

$$T(n/a) = T(n/a^2) + n/2$$

$$T(n) = [T(n/2^2) + \frac{n}{2}] + n$$

$$= \underbrace{T(n/2^2)}_{\downarrow} + n/2 + n$$

$$= [T(n/2^3) + n/2^2] + \frac{n}{2} + n$$

$$= T(n/2^k) + n/2^{k-1} + n/2^{k-2} + \dots + n/2^1 + n$$

$$\therefore \text{Again assume } n/2^k = 1$$

$$k = \log n$$

$$T(n) = 1 + n[1 + 1]$$

$$= 1 + 2n$$

$$\boxed{O(n)}$$